

Fact-Checking Pipeline

Design, Implementation & Early Evaluation

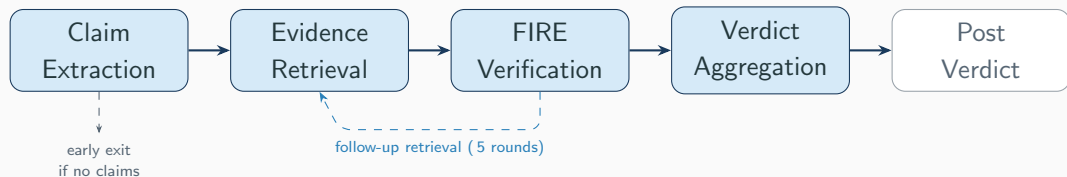
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What We Are Building

A **browser extension** that intercepts social media posts *before sharing* and issues a misinformation nudge if the content is likely false or misleading.

- **Platforms:** Twitter/X and Facebook
- **Languages:** English + French
- **Research arm:** Empirical validation of the Guriev et al. persuasion model — anonymised behavioural data collected at scale
- **Key constraint:** User-facing latency under one second; verification is dispatched asynchronously

Pipeline Overview



Model	Sources	Model	Priority rule
Qwen3-32B Groq	FCTAPI · Wiki Serper + scrape	Llama-3.3-70B Groq	Refuted > CE > NEI > Supp.

- **Is this post factually checkable?**

Decided by *form*, not plausibility. “The earth is flat” is checkable. Opinions, predictions, and questions are not.

- **Decompose into atomic claims**

Each claim is a single, independently verifiable assertion.

- **Generate 3 search queries per claim**

Direct factual · contextual · debunking-oriented

- **Early exit** if no checkable claims found

5–8 **analytical questions** per claim — the interpretive scaffold the verifier reasons over in Step 3. Not search queries.

- Official statements and attributions
- Expert or scientific consensus
- Contradicting sources
- Quantitative specifics — numbers, dates, scope
- Temporal anchoring

- **Google Fact Check Tools API** cred = 0.90
Structured verdicts from AFP, PolitiFact, Snopes, and hundreds of regional outlets
- **Wikipedia** cred = 0.75
Top-2 article summaries for named entities and events in the claim
- **Serper web search** cred from CRED-1
3 queries $\times$ 3 URLs, date-capped to the claim date
Filtered: fake/unreliable domains ($\text{CRED-1} < 0.30$) and structural blocklist
(paywalls, login walls)

Step 2 — Filtering & Ranking

After scraping, all evidence from the three streams is merged and processed:

- **BM25 chunk extraction**

Top-2 × 300-word chunks per scraped page, scored against the claim text

- **Cross-encoder reranking**

ms-marco-MiniLM-L6-v2

All items scored versus the claim; top 10 retained

- **MMR diversification**

$\lambda = 0.75$ relevance / 0.25 diversity

Selects up to 5 final items; prevents the verifier from receiving multiple paraphrases of the same source

- Answers each verifying question from evidence, citing source URLs
- Weights higher-credibility sources (≥ 0.6) more heavily
- Rates **all four labels independently** on a 1–5 Likert scale:
Supported · Refuted · NEI · Conflicting Evidence
- Verdict = label with highest Likert score
Ties: Refuted > Supported > CE > NEI
- Returns reasoning trace, one-sentence justification, and key URLs

Step 3 — FIRE Iterative Loop

After each verification round:

- Compute **label margin** = top Likert score – second-highest
- **Exit immediately** if verdict $\neq$ NEI and margin ≥ 4
- **Always continue** if verdict = NEI — model declared insufficient evidence
- **Stop** if follow-up query is redundant (cosine sim > 0.9 vs. all past queries)
- Otherwise: generate targeted follow-up query, retrieve, merge, and repeat

Maximum 5 rounds.

Likert Rubric — Supported & Refuted

Supported — “evidence confirms the claim”

- 5 Two+ credible sources (≥ 0.6) confirm all key qualifiers; no credible contradiction
- 4 Multiple sources agree, or one high-cred (≥ 0.7) source confirms directly
- 3 One credible (≥ 0.5) source supports; no corroboration
- 1–2 Weak or off-topic support only

Refuted — “evidence contradicts the claim”

- 5 Two+ credible sources (≥ 0.6) contradict a load-bearing qualifier; no credible support
- 4 One high-cred (≥ 0.7) source directly refutes a key element
- 3 One credible (≥ 0.5) source contradicts; no corroboration
- 1–2 Weak or off-topic contradiction only

Likert Rubric — NEI & Conflicting Evidence

Not Enough Evidence

“evidence does not address the claim”

- 5** No retrieved item touches the claim at all
- 3–4** Topically related sources only; no direct address
- 1–2** At least one credible source (≥ 0.5) speaks to the substance — *must not* score NEI above 2 in this case

NEI is about coverage, not certainty.

Conflicting Evidence

“credible sources meaningfully disagree”

- 5** Two+ high-cred (≥ 0.6) sources on each side
- 4** One credible source supports AND one credible source refutes the same load-bearing element
- 1–3** All credible sources agree, or only minor wording differences

If Supported and Refuted both ≥ 3 , raise CE to at least $\max(\text{Supp}, \text{Ref})$.

ClaimCheck arXiv:2510.01226 · NAACL 2025

Multi-hop QA generation: 5–8 targeted verifying questions before retrieval. Qwen3-4B achieves **76.4%** on AVeriTeC dev, outperforming GPT-4o. Our pipeline scores $\approx 60\%$. Gap is architecture, not model size.

→ Verifying-question generation (Step 1)

FIRE arXiv:2411.00784 · NAACL 2025 Findings

Confidence-gated iterative retrieval with early exit. Comparable accuracy with **7.6× fewer LLM calls** and **16.5× fewer search calls**.

→ FIRE loop (Step 3)

HerO 2 arXiv:2507.11004 · AVeriTeC 2025

Per-document summarisation and answer reformulation before verification.

Enforces temporal anchoring on evidence text — prevents the model from substituting parametric knowledge for period-appropriate sources.

→ Planned next iteration (targets date-sensitivity failure)

CTU AIC arXiv:2508.04390 · AVeriTeC 2025 winner

Likert per-label confidence scoring and MMR diversity reranking. Ablation: thinking tokens provide *no accuracy gain* — token budget better spent on retrieval coverage.

→ Likert scoring & MMR diversification (Steps 2–3)

Misleadingness Detection — The Problem

All SOTA systems target *stated factual claims*: is what the post says true? None detect posts that are **technically true but misleading**.

- **Misleading by scope**

A broad claim backed only by a narrow instance.

“A study shows vaccines cause autism” — one retracted paper, not evidence.

- **Misleading by implication**

A true statement that naturally suggests a false conclusion.

“Cities with more immigrants have higher crime rates” — implies causation; omits confounders.

- **Misleading by omission**

True only under unstated conditions.

Misleadingness Detection — The Algorithm

Applied after explicit claim verification. For each verified claim A:

1. Establish A is true, preserving all qualifiers (time period, sample size, scope)
2. **Strip qualifiers** $\rightarrow \tilde{A}$. If $\tilde{A}$ is false $\rightarrow$ [misleading by scope](#)
3. **Derive implication** B: the first factual inference a reasonable person draws from A. If B is false $\rightarrow$ [misleading by implication](#)
4. **Augment with silent assumptions** $\rightarrow A^+$. If A^+ is false $\rightarrow$ [misleading by omission](#)

Each derived claim is fed through the standard retrieval + verification sub-pipeline. A misleading verdict raises the post's falsity score even when the literal claim is *Supported*.

Evaluation — AVeriTeC Benchmark (10-claim sample)

#	Claim (abbreviated)	Gold	Pipeline	
1	Trump canceled HR6666	Ref.	Ref.	✓
2	US–Africa trade declined	Supp.	NEI	✗
3	Chakraborty father statement	Ref.	NEI	✗
4	Fox News unavailable in Canada	Ref.	Ref.	✓
5	Nigeria 52% urban	Supp.	Ref.	✗
6	Biden transgender quote	NEI	Ref.	✗ [†]
7	RBG age-of-consent claim	Ref.	Ref.	✓
8	Trump canceled ID2020	Ref.	Ref.	✓
9	Thackeray / Kangana (compound)	Ref.	Ref.	✓
10	Schools reopened (New Brunswick)	Supp.	NEI	✗

Evaluation — Error Analysis

Of the 5 misses:

- **Retrieval coverage gaps** (claims 3, 10)

Regional or niche claims with no English web coverage. NEI is correct given what retrieval returned — the failure is upstream.

- **Date sensitivity** (claim 5)

Verifier retrieved 2024 statistics for a 2020 claim and used parametric knowledge instead of period-appropriate evidence.

Fix: [HerO 2 per-document summarisation with temporal anchoring](#).

- **Annotation disagreement** (claim 6)

Four professional fact-checkers rated it Refuted; AVeriTeC gold = NEI on stricter wording. Genuine label conflict, not a pipeline error.

Evaluation — Larger-Scale Runs

Run	<i>n</i>	Accuracy
FIRE baseline (seed 42)	50	60%
SC ×3 (seed 0)	200	61%

Label (SC, <i>n</i> =200)	Gold	Recall
Refuted	123	67%
Supported	55	62%
NEI	10	60%
CE	12	0%

- SC lifted Supported recall **+19 pp** (43% → 62%)
- Refuted recall dropped slightly (71% → 67%)
- Net accuracy unchanged — runs use different seeds
- ECE = 0.35: mean confidence ≈ 0.95 vs. accuracy = 0.61 — severely overconfident
- Dominant failure: NEI over-prediction
57/200 (28.5%) predicted NEI vs. 5%

Conflicting Evidence Recall = 0%

12 gold-CE claims in the 200-claim run. Predicted: 6 Refuted, 5 NEI, 1 Supported, 0 CE.

Under self-consistency ($\times 3$ runs), averaging Likert scores across rounds systematically suppresses CE: if runs 1 and 3 lean Refuted and run 2 leans CE, the averaged CE score rarely becomes the argmax.

Fix candidates:

- **Majority-vote aggregation** — one-line change; test this first
- **Targeted CE retrieval** — after Supported/Refuted, generate a query explicitly searching for the opposing view
- **Separate CE classifier** — fine-tuned on evidence pairs; bypasses LLM CE-avoidance bias

Scenario A — Experimentation Phase: Cost

One-time experiment: 1 000 users × 100 posts = **100 000 pipeline calls.**

Tokens per check

Component	Tokens
Extraction (Qwen3-32B)	~1 700
FIRE (Llama-3.3-70B)	~14 400
Total / check	~16 100

Experiment cost

Item	Total
Qwen3-32B	\$119
Llama-3.3-70B	\$994
Serper	\$700
Hosting	\$50
One run	~\$1 900
3-config comparison	~\$4 000

Scenario A — Model Cost Comparison (100 K checks)

Grand total = extraction + verification + \$750 (Serper \$700 + hosting \$50).

Extraction cost

Model	Cost
Llama-3.1-8B	\$9
GPT OSS 20B	\$21
Llama-4-Scout	\$27
GPT OSS 120B	\$42
Qwen3-32B	\$59
Llama-3.3-70B	\$104

Verification cost

Model	Cost	Eval
GPT OSS 20B	\$148	untested
Llama-4-Scout	\$197	~40% ($n=10$)
GPT OSS 120B	\$297	untested
Qwen3-32B	\$456	untested
Llama-3.3-70B	\$840	61% ($n=200$, $SC\times 3$)

Example totals: Qwen3-32B + Llama-3.3-70B (*current*): \$1,649 Llama-3.1-8B + GPT OSS ^{20/24}

Scenario A — Stack

Stack — cloud APIs only

- Extraction: Qwen3-32B via Groq
- Verification: Llama-3.3-70B via Groq, full FIRE loop
- Retrieval: BM25 + cross-encoder + MMR
- Web search: Serper API
- Fact-check KB: Google FCTAPI (free)
- Backend: single FastAPI server (~\$50)

Model comparison strategy

Run retrieval once per claim, cache the evidence set, then re-run only the verifier with different model configs. Each additional config adds ~\$1 100.

What we skip for now: vector DB, async queue, French language support

Timeline: 1–2 engineers · 2–4 weeks

Cost Lever: KB Routing Before Web Search

Currently all three sources (FCTAPI, Wikipedia, Serper) run in parallel on every claim. At scale, this means paying for Serper queries on claims professional fact-checkers have already verified.

Fix: two-phase sequential routing

1. Query FCTAPI + vector DB first (free, ~300 ms)
2. KB hit → return verdict, skip Serper, FIRE exits after round 1
3. No match → proceed to full Serper retrieval + FIRE

Expected impact: ~40% of novel claims resolved by KB in a mature system. KB hits carry credibility 0.90 — FIRE almost always exits after round 1.

Code change: parallel `ThreadPoolExecutor` in `retrieval.py` becomes two-phase.

Scenario B — Wide Study: Cost

Recurring: 20 000 users × 100 posts/day = **2M pipeline calls/day**.

LLM + search at API pricing				Team	
Optimisation	Calls/day	/day	/month	Role	
None (raw)	2M	\$35 500	\$1.1M	Backend + infra	2–3
Claim cache (80% hit)	400K	\$7 100	\$213K	Browser ext.	1
Cache + KB routing	240K	\$5 050	\$152K	ML / eval	1
				PM / IRB	1

All three scenarios require self-hosted HPC.

Compute drops 6× from raw to optimised.

Scenario B — Wide Study: Stack & Compute

Compute path

- **HPC allocation** — DRAC (Canada), Jean Zay (France)
GPU-hours at zero marginal cost for academic research. [Recommended path](#).
- **Cloud GPU cluster** — \$30–50K/month
~50–80× A100 80 GB with vLLM
- **Smaller model** — 7B params cuts GPU needs $\approx 10\times$ at some accuracy cost

New infrastructure vs. Scenario A

- **Qdrant** — vector DB for claim cache
- **Redis + Celery** — async Tier 3 job queue
- **Load balancer** — 10-country deployment
- **Observability** — latency per tier, nudge rate, cache hit rate

Timeline to production: 4–6 months